

INTUITION_OS_DATABASE_SCHEMA_V2.md

Version 1.0

DESIGN PRINCIPLES

The database stores:

- Facts
- Events
- Evidence

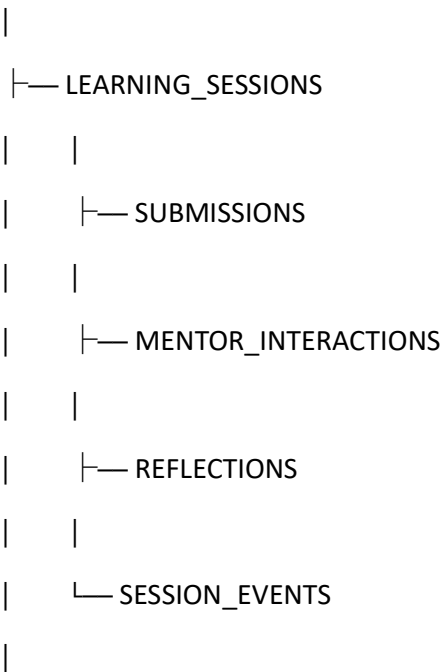
The database does NOT store:

- AI opinions
- Personality labels
- Permanent strengths
- Permanent weaknesses

All insights are generated dynamically.

ENTITY RELATIONSHIP DIAGRAM

USERS



└─ EVIDENCE

USERS

Purpose:

Identity and authentication.

CREATE TABLE users (

id UUID PRIMARY KEY,

email VARCHAR(255) UNIQUE NOT NULL,

name VARCHAR(255) NOT NULL,

avatar_url TEXT,

provider VARCHAR(50) NOT NULL DEFAULT 'google',

provider_id VARCHAR(255) NOT NULL,

created_at TIMESTAMP NOT NULL DEFAULT NOW(),

updated_at TIMESTAMP NOT NULL DEFAULT NOW()

);

Indexes:

CREATE UNIQUE INDEX idx_users_email

ON users(email);

LEARNING_SESSIONS

Core domain object.

Every learning activity belongs to a session.

```
CREATE TABLE learning_sessions (  
  id UUID PRIMARY KEY,  
  
  user_id UUID NOT NULL REFERENCES users(id),  
  
  problem_title TEXT NOT NULL,  
  
  problem_url TEXT NOT NULL,  
  
  difficulty VARCHAR(20),  
  
  tags JSONB DEFAULT '[]',  
  
  status VARCHAR(20) NOT NULL,  
  
  started_at TIMESTAMP NOT NULL,  
  
  ended_at TIMESTAMP,  
  
  duration_seconds INTEGER,  
  
  total_submissions INTEGER DEFAULT 0,  
  
  mentor_requests INTEGER DEFAULT 0,  
  
  highest_escalation_level INTEGER DEFAULT 0,
```

editorial_viewed BOOLEAN DEFAULT FALSE,

solved BOOLEAN DEFAULT FALSE,

created_at TIMESTAMP DEFAULT NOW()

);

Status:

ACTIVE

COMPLETED

ABANDONED

Indexes:

CREATE INDEX idx_sessions_user

ON learning_sessions(user_id);

CREATE INDEX idx_sessions_started

ON learning_sessions(started_at DESC);

SUBMISSIONS

Tracks all submission outcomes.

CREATE TABLE submissions (

id UUID PRIMARY KEY,

session_id UUID NOT NULL

REFERENCES learning_sessions(id)

ON DELETE CASCADE,

result VARCHAR(20) NOT NULL,

```
    created_at TIMESTAMP NOT NULL DEFAULT NOW()
);
```

Results:

WA

TLE

MLE

RE

AC

Indexes:

```
CREATE INDEX idx_submissions_session
```

```
ON submissions(session_id);
```

MENTOR_INTERACTIONS

Stores every mentor exchange.

```
CREATE TABLE mentor_interactions (
    id UUID PRIMARY KEY,

    session_id UUID NOT NULL
        REFERENCES learning_sessions(id)
        ON DELETE CASCADE,

    escalation_level INTEGER NOT NULL,

    observation TEXT,

    question TEXT,

    guidance TEXT,
```

```
    user_response TEXT,  
  
    created_at TIMESTAMP DEFAULT NOW()  
);
```

Indexes:

```
CREATE INDEX idx_mentor_session  
ON mentor_interactions(session_id);
```

REFLECTIONS

One reflection per session.

```
CREATE TABLE reflections (  
    id UUID PRIMARY KEY,  
  
    session_id UUID UNIQUE NOT NULL  
        REFERENCES learning_sessions(id)  
        ON DELETE CASCADE,  
  
    session_summary TEXT,  
  
    learning_insight TEXT,  
  
    growth_impact TEXT,  
  
    next_step TEXT,  
  
    raw_json JSONB,
```

```
created_at TIMESTAMP DEFAULT NOW()  
);
```

EVIDENCE

Most important table.

Stores objective learning evidence.

```
CREATE TABLE evidence (  
  id UUID PRIMARY KEY,  
  
  user_id UUID NOT NULL  
    REFERENCES users(id),  
  
  session_id UUID  
    REFERENCES learning_sessions(id),  
  
  evidence_type VARCHAR(100) NOT NULL,  
  
  confidence VARCHAR(20) NOT NULL,  
  
  metadata JSONB DEFAULT '{}',  
  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

Examples:

SOLVED_WITHOUT_HINT

SOLVED_WITHOUT_EDITORIAL

RECOVERED_FROM_WA

RECOVERED_FROM_TLE

PATTERN_RECOGNIZED

PATTERN_MISSED

OPTIMIZATION_DISCOVERED

EDGE_CASE_FIXED

MULTIPLE_FAILED_ATTEMPTS

NEEDED_HINT_LEVEL_1

NEEDED_HINT_LEVEL_2

NEEDED_HINT_LEVEL_3

NEEDED_HINT_LEVEL_4

USED_EDITORIAL

Indexes:

CREATE INDEX idx_evidence_user

ON evidence(user_id);

CREATE INDEX idx_evidence_type

ON evidence(evidence_type);

SESSION_EVENTS

Critical for future intelligence.

Stores raw events.

Instead of generating evidence directly from sessions,
we store event history.

```
CREATE TABLE session_events (  
    id UUID PRIMARY KEY,  
  
    session_id UUID NOT NULL  
        REFERENCES learning_sessions(id)  
        ON DELETE CASCADE,  
  
    event_type VARCHAR(100) NOT NULL,  
  
    payload JSONB DEFAULT '{}',  
  
    created_at TIMESTAMP DEFAULT NOW()  
);
```

Examples:

PROBLEM_OPENED

HELP_REQUESTED

MENTOR_LEVEL_1

MENTOR_LEVEL_2

SUBMISSION_WA

SUBMISSION_TLE

SUBMISSION_AC

EDITORIAL_VIEWED

SESSION_COMPLETED

Why?

Because future versions can re-process events
and generate better evidence.

USER_SETTINGS

Minimal preferences table.

```
CREATE TABLE user_settings (
```

```
    user_id UUID PRIMARY KEY
```

```
    REFERENCES users(id)
```

```
    ON DELETE CASCADE,
```

```
    preferred_ai_provider VARCHAR(50),
```

```
    onboarding_completed BOOLEAN DEFAULT FALSE,
```

```
    created_at TIMESTAMP DEFAULT NOW()
```

```
);
```

WHAT WE INTENTIONALLY DO NOT STORE

Do NOT store:

strength_score

dp_score

graph_score

intelligence_score

confidence_score

resilience_score

skill_score

Reason:

These become stale.

Generate them dynamically from evidence.

PROFILE GENERATION QUERY STRATEGY

Profile is generated from:

Last 90 Days Evidence

-

Recent Sessions

-

Topic History

Nothing is permanently stored.

MVP DATABASE SIZE ESTIMATE

1 user solving:

3 problems/day

For 1 year:

~1000 sessions

~5000 events

~3000 evidence records

PostgreSQL handles this easily.

No scaling concerns in V1.

FUTURE TABLES (NOT MVP)

weekly_reviews

monthly_reviews

recommendations

learning_paths

topic_progress

intervention_history

Do not build initially.

FINAL DATABASE RULE

Events are truth.

Evidence is derived.

Insights are generated.

Profiles are assembled.

Never reverse this pipeline.

EVENTS



EVIDENCE



INSIGHTS



PROFILE



RECOMMENDATIONS

END OF DOCUMENT